LAB 04

Hypothesis Stage: from a Written Card to an Automated Statistical Test

Item	Detail
Topic	Topic 1: Foundations of Quant and Algorithmic Trading: From Thorp to AI Agents
Pipeline stage	Hypothesis
TSC reference	A1, A6, K1
Duration	15 min
Folder	labs/Lab04-trading-hypothesis/
Files	Lab04-trading-hypothesis/hypothesis_stage.py, pipeline/hypotheses/spy_dip_after_streak.yaml, pipeline/hypothesis.py, Lab04-trading-hypothesis/solution/hypothesis.py, Lab04-trading-hypothesis/solution/spy_dip_after_streak.yaml
Simulator page	http://localhost:8000/#/research/hypothesis
Mode	Runs ONLINE with your Alpaca paper keys (.env), or OFFLINE on the bundled data and the paper-broker simulator. Paper trading only.

Lion City Capital and every scenario in these materials are fictitious and for training only. All trading in this course is paper (simulated) trading. Nothing in these materials is investment advice or a recommendation to trade any security.

Scenario

A portfolio manager believes SPY tends to bounce after several losing days. The desk's rule is that no idea enters the signal and backtest stages until it is written as a hypothesis card - before anyone looks at results - and passes an automated statistical test with a decision rule that includes trading costs.

Objective

Write a falsifiable hypothesis card in YAML (H1, H0, signal known at the close, outcome, test, decision rule, big-data use case) and build `python -m pipeline hypothesis` to test it with a Welch t-test, a bootstrap confidence interval and a robustness grid.

Before You Start

- Part A: start the Trading Simulator (uv run trader serve --with-worker, or docker compose up -d) and open http://localhost:8000 > Course Labs > Lab 04. Install steps: Learner Guide, 'Download, Install and Start the Trading Simulator Software'.
- Part B: open a terminal in the labs folder, activate your virtual environment and open your AI coding assistant there; the finished reference code is already in this folder.
- Record every result you are asked to check - the same evidence is asked for in the Practical Performance assessment.

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Part A - In the Trading Simulator

Start the simulator (uv run trader serve --with-worker, or docker compose up -d) and open http://localhost:8000. In the sidebar open Course Labs > Lab 04: the app shows the same steps with a Go button that opens each page pre-filled. Every step works offline on the bundled demo data and the built-in paper broker.

A1. Write the hypothesis card

Research > Hypothesis test. Keep signal 'losing streak' n = 3, horizon 1 day, alpha 0.05, min edge 3 bps. Read the H1 and H0 text.

Open: http://localhost:8000/#/research/hypothesis (Research > Hypothesis (SPY, losing streak 3))

Check: H0 and H1 are written before you press Run.

A2. Run the test

Press 'Run the test'.

Open: http://localhost:8000/#/research/hypothesis

Check: You can read the signal-day mean, the other-day mean, Welch t and p, and the 95% bootstrap CI.

A3. Robustness and multiple testing

Read the robustness grid (other streak lengths and QQQ) and the Bonferroni alpha.

Open: http://localhost:8000/#/research/hypothesis

```
Command line: from trader.core.stats import hypothesis_test
# hypothesis_test(bars, {'type': 'losing_streak', 'n': 3}, horizon=1)
```

Check: You can explain why trying many variants inflates false positives.

Expected in the app: A decision (SUPPORTED / NOT SUPPORTED) with p-value, edge in bps and a robustness grid.

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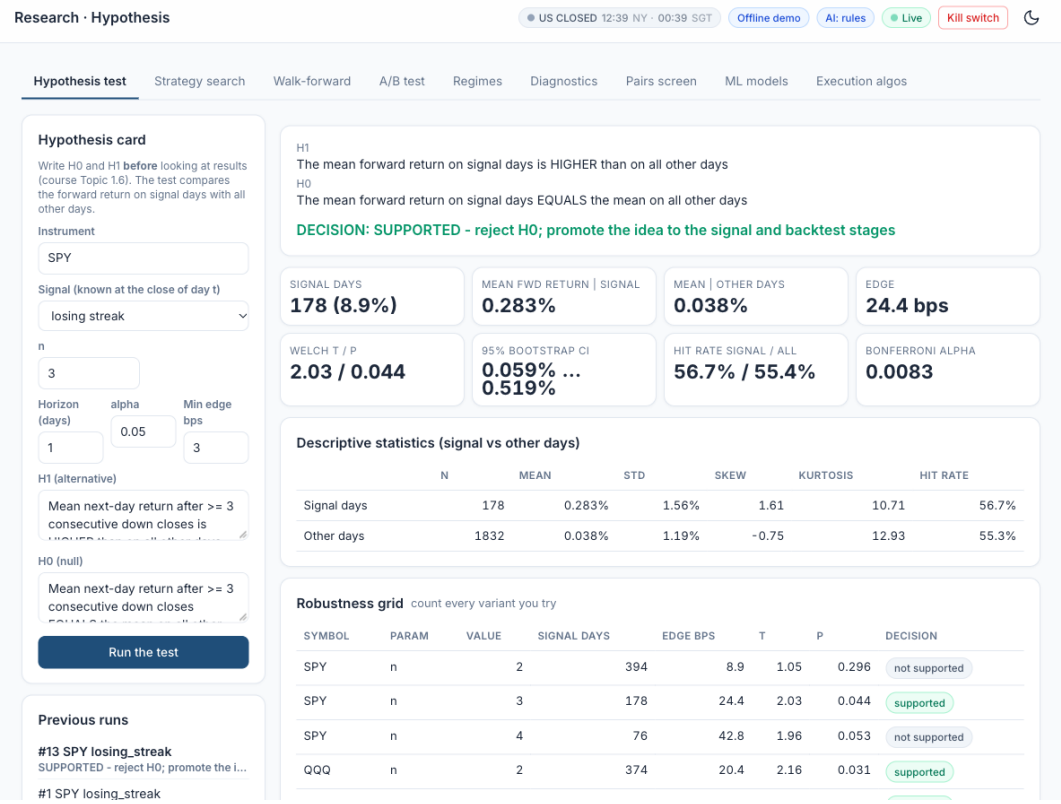


Figure: Lab 04 in the Trading Simulator (Lab04_hypothesis.png).

Part B - In Code with Your AI Assistant

The same pipeline stage in code. AI writes the code; you own the algorithm: paste each prompt into your AI coding assistant in the labs folder (the finished reference code is already in labs/Lab04-trading-hypothesis/), run it, and check the result against Part A. The Practical Performance assessment uses this code path.

Step 1: Read the hypothesis card (A1, K1)

Open pipeline/hypotheses/spy_dip_after_streak.yaml. Identify H1, H0, the signal (3 consecutive down closes, known at the close), the outcome (next-day return), the test and the decision rule ($p < 0.05$ AND an edge of at least 3 bps, the estimated round-trip cost).

Check: You can say why the outcome must be a FUTURE return and why the decision rule includes a cost hurdle.

Step 2: Write a card for your own idea (A1, K1)

Copy the card to pipeline/hypotheses/my_card.yaml and adapt it to an idea of your own (for example QQQ after closing below its 200-day average, or AAPL after a losing streak). Write it BEFORE running anything.

AI prompt - paste into your AI coding assistant
Help me write a hypothesis card in YAML for this trading idea: <describe your idea in one sentence>. Use exactly the fields of labs/pipeline/hypotheses/spy_dip_after_streak.yaml (id, name, instrument, timeframe, sample, rationale, h1, h0, signal, outcome, test, decision_rule, robustness, biases_to_watch, big_data_use_case). The signal type must be one of losing_streak (n), rsi_below (window, level) or close_below_sma (window) and must be known at the close of day t; the outcome must be a forward return over horizon_days; H0 must be 'no difference'. Do not analyse any data.

Check: my_card.yaml has an economic rationale, H1, H0, a signal known at the close and a decision rule.

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Step 3: Build the test engine (A1)

Ask your assistant for the core of pipeline/hypothesis.py.

AI prompt - paste into your AI coding assistant
Create labs/pipeline/hypothesis.py. losing_streak(close) counts consecutive down closes ending at each day. condition(bars, signal) builds the boolean signal for the types losing_streak, rsi_below (RSI from common.indicators) and close_below_sma, using data up to the close of day t only. test_card(bars, card, signal=None): outcome fwd = close.shift(-horizon) / close - 1 (the outcome, never a feature); compare signal days with all other days: counts, means, hit rates, scipy.stats.ttest_ind(equal_var=False), a bootstrap 95% CI of the signal-day mean with numpy default_rng(card seed) and card bootstrap draws, edge_bps = (mean signal - mean other) x 10,000, supported = p < alpha AND t > 0 AND edge_bps >= min_edge_bps. Load bars with pipeline.ingest.load_bars. Rules: run from the labs folder; reuse labs/common (config, data, broker, indicators, metrics, backtest, plotting) and earlier pipeline stages instead of rewriting them; do not modify common/ or data/; read every setting from pipeline/config.yaml via pipeline.config.load_config(); never put API keys or secrets in code; paper trading only; keep a main(argv) function so `python -m pipeline <command>` works; add short comments a finance professional can follow.

Check: test_card returns a dict with signal_days, means, ci95, welch_t, p_value, edge_bps and supported.

Step 4: Add the robustness grid, chart and result file (A1, A6)

Ask your assistant for run() and main().

AI prompt - paste into your AI coding assistant
In labs/pipeline/hypothesis.py add run(card_path=None) for `python -m pipeline hypothesis [--card PATH]` (default pipeline/hypotheses/spy_dip_after_streak.yaml): print H1, H0, sample, signal days, means, hit rates, CI, Welch t-test, edge and a DECISION line ('SUPPORTED - reject H0; promote the idea to the signal and backtest stages' or 'NOT SUPPORTED - do not reject H0'); a robustness grid over card.robustness.streaks for the instrument and each robustness symbol with a Bonferroni threshold alpha / number of variants; a table of forward return by streak length (0..4, 5+) with standard errors; chart Lab04_hypothesis.png (bars with 95% error bars, unconditional mean line); and write the card plus results to _outputs/Lab04_hypothesis_result.yaml. Rules: run from the labs folder; reuse labs/common (config, data, broker, indicators, metrics, backtest, plotting) and earlier pipeline stages instead of rewriting them; do not modify common/ or data/; read every setting from pipeline/config.yaml via pipeline.config.load_config(); never put API keys or secrets in code; paper trading only; keep a main(argv) function so `python -m pipeline <command>` works; add short comments a finance professional can follow.

Check: python -m pipeline hypothesis --help shows --card.

Step 5: Run the test on the desk's card (A1)

Run: python -m pipeline hypothesis (same as: python3 Lab04-trading-hypothesis/hypothesis_stage.py).

Check: SPY 2018-01-02 to 2025-12-30 (2,010 days): 178 signal days (8.9%); mean next-day return +0.2828% vs +0.0384% on other days (unconditional +0.0601%); hit rate 56.7% vs 55.4%; bootstrap CI [+0.0531%, +0.5136%]; Welch t = 2.03, p = 0.044; edge +24.4 bps; 'DECISION: SUPPORTED'.

Step 6: Read the robustness grid (A1, K1)

Read the robustness table and the Bonferroni line. Which variants would survive if the desk counted every test it ran?

Check: 3 of 6 variants supported: SPY n=3 (p 0.044), QQQ n=2 (p 0.031) and QQQ n=3 (p 0.006); SPY n=2 (p 0.296), SPY n=4 (p 0.053) and QQQ n=4 are not. Bonferroni threshold p < 0.0083 - only QQQ n=3 passes it.

Step 7: Study the shape of the effect (A1)

Read the table 'Forward return by losing-streak length' and open _outputs/Lab04_hypothesis.png.

Check: Streaks of 4 (49 days, +0.439%) and 5+ (27 days, +0.533%) look strongest but have the widest error bars - small samples.

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Step 8: Test your own card and its big-data use case (A1, A6)

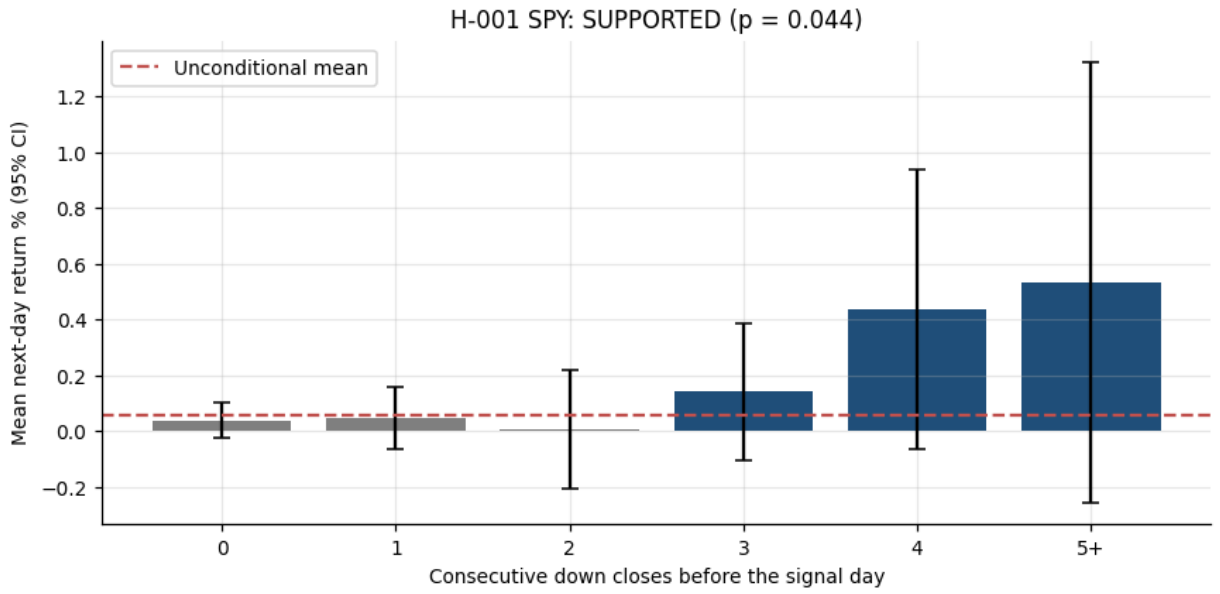
Run: python -m pipeline hypothesis --card pipeline/hypotheses/my_card.yaml. Then complete the big_data_use_case section of your card (data, source, tooling, governance, success measure).

AI prompt - paste into your AI coding assistant
Review my hypothesis result in labs/_outputs/Lab04_hypothesis_result.yaml. In plain English, say whether the evidence supports H1, how many variants I tested, whether the result survives a Bonferroni correction, and which bigger or alternative data set (volume, velocity or variety) would test the idea better. Do not change any code.

Check: Your card prints a DECISION line and _outputs/Lab04_hypothesis_result.yaml is rewritten for your card.

Expected Output

python -m pipeline hypothesis: SPY streak >= 3 - 178 signal days (8.9%), mean next-day return +0.2828% vs +0.0384% (all days +0.0601%), hit rate 56.7% vs 55.4%, bootstrap CI [+0.0531%, +0.5136%], Welch t = 2.03, p = 0.044, edge +24.4 bps -> SUPPORTED; robustness 3 of 6 variants supported, Bonferroni p < 0.0083; files _outputs/Lab04_hypothesis_result.yaml and chart _outputs/Lab04_hypothesis.png.



Lab 4 output (Lab04_hypothesis.png), offline mode on the bundled data.

Debrief

Writing the hypothesis before testing, with a null, a cost-aware decision rule and a count of the variants tried, is the foundation of data-driven trading (A1). AI and machine-learning methods (K1) only add value when they test well-posed questions, and larger data sets (A6) are how a promising effect in one ETF is confirmed or rejected.

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Troubleshooting

Symptom	What to do
p-value changes when you change the dates	Expected - small samples are sensitive. Record it as evidence, not as a problem to hide.
ValueError: unknown signal type	Use losing_streak, rsi_below or close_below_sma, or ask your assistant to add a new type to condition().
Very few signal days	With fewer than about 30 observations the test has little power; loosen the condition.
Unsure what H0 should be	H0 is 'no difference': the mean forward return on signal days equals the mean on other days.

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